Follow Up Questions to test Arithmetic accuracy

Q1. Company named ABC Ltd has a lease for 05 years with a monthly payment of $8000 which is paid in advance. The current discount rate applied is 5% and the lease start date is 01st of January 2025 onwards. Calculate the Lease Liability: Present value of monthly payments over 5 years and the Right-of-Use Asset: Equals lease liability at commencement.

*** MY CONCERN SAME QUESTION GAVE ME TWO ANSWER IN THREE DIFFERENT TIME LINES IS THAT POSSIBLE – Examples Below

ANSWER AT 10.29 AM – ON CRUNCH

Calculation of Lease Liability and Right-of-Use Asset

To calculate the lease liability and the right-of-use asset for ABC Ltd, we need to determine the present value of the monthly lease payments over the lease term of 5 years.

Steps to Calculate Lease Liability

1. **Identify the parameters:**
    - Monthly payment (PMT) = A$8,000
    - Lease term = 5 years
    - Number of payments = 5 years * 12 months/year = 60 months
    - Discount rate (r) = 5% per annum / 12 months = 0.41667% per month
2. **Calculate the present value of the lease payments:** The formula for the present value of an annuity is:

PV = PMT * [(1 - (1 + r)^-n) / r]

Where:

- PV = Present Value
- PMT = Payment per period
- r = Interest rate per period
- n = Total number of payments

3. **Substituting the values:**

PV = 8000 * [(1 - (1 + 0.0041667)^-60) / 0.0041667]

4. **Calculate the present value:**
5. PV = 8000 * [(1 - (1.0041667)^-60) / 0.0041667]
6. PV = 8000 * [0.776839 / 0.0041667]
7. PV = 8000 * 186.078

PV = 1,488,624

8. **Lease Liability:** The lease liability is the present value of the future lease payments, which is approximately **A$1,488,624**.

Right-of-Use Asset

According to AASB 16, the right-of-use asset is initially measured at the amount of the lease liability. Therefore, the right-of-use asset for ABC Ltd is also **A$1,488,624**.

Summary of Calculations

- **Lease Liability:** A$1,488,624
- **Right-of-Use Asset:** A$1,488,624

Answer at 1.59pm on CRUNCH

Calculation of Lease Liability and Right-of-Use Asset

To calculate the lease liability and right-of-use asset for ABC Ltd, we need to determine the present value of the lease payments over the 5-year period.

Step 1: Calculate the Present Value of Lease Payments

1. **Monthly Payment**: A$8,000
2. **Number of Payments**: 5 years * 12 months = 60 payments
3. **Discount Rate**: 5% per annum (or approximately 0.4167% per month)

The present value (PV) of an annuity formula is used here:

PV = Pmt × [(1 - (1 + r)^-n) / r]

Where:

- Pmt = monthly payment (A$8,000)
- r = monthly discount rate (5% / 12 months = 0.004167)
- n = total number of payments (60)

Calculating the present value:

Pmt = 8000

r = 0.004167

n = 60

PV = Pmt × [(1 - (1 + r)^-n) / r]

PV = 8000 × [(1 - (1 + 0.004167)^-60) / 0.004167]

PV = 8000 × [1 - (1.004167)^-60] / 0.004167

PV = 8000 × [1 - 0.7792] / 0.004167

PV = 8000 × 52.195

PV ≈ 417,560

Step 2: Recognize the Lease Liability and Right-of-Use Asset

- **Lease Liability**: A$417,560 (the present value of future lease payments)
- **Right-of-Use Asset**: Equals lease liability at commencement = A$417,560

Summary of Calculations

Item	Amount (AUD)
Lease Liability	A$417,560
Right-of-Use Asset	A$417,560

ABC Ltd will recognize a lease liability of A$417,560 and a right-of-use asset of A$417,560 at the commencement of the lease on January 1, 2025.

Answer at 2.08 pm – On Crunch

Calculation of Lease Liability and Right-of-Use Asset for ABC Ltd

To calculate the lease liability and the right-of-use asset for ABC Ltd, we will follow these steps:

1. **Determine the total number of payments:** Since the lease is for 5 years with monthly payments, the total number of payments will be:
    - Total Payments = 5 years * 12 months/year = 60 months
2. **Calculate the present value of the lease payments**: The monthly payment is A$8,000, and the discount rate is 5% per annum. The present value of an annuity formula will be used for this calculation.
3. **Calculate the right-of-use asset**: The right-of-use asset is equal to the lease liability at commencement.

Present Value Calculation

The present value of an annuity can be calculated using the formula:

PV = Pmt * [(1 - (1 + r)^-n) / r]

Where:

- PV = Present Value
- Pmt = Payment per period (A$8,000)
- r = Discount rate per period (5% per annum / 12 months = 0.41667% per month)
- n = Total number of payments (60)

Calculation Steps

1. **Convert the annual discount rate to a monthly rate**:
    - Monthly discount rate = 5% / 12 = 0.41667% = 0.0041667
2. **Substitute the values into the formula**:

Pmt = 8000

r = 0.0041667

n = 60

PV = 8000 * [(1 - (1 + 0.0041667)^-60) / 0.0041667]

Final Calculation

Let's calculate the present value:

PV = 8000 * [(1 - (1 + 0.0041667)^-60) / 0.0041667]

PV = 8000 * [1 - (1.0041667)^-60] / 0.0041667

PV = 8000 * [1 - 0.7792] / 0.0041667

PV = 8000 * 52.258

PV = 418064.00

Summary of Results

- **Lease Liability**: A$418,064.00
- **Right-of-Use Asset**: A$418,064.00 (equal to the lease liability at commencement)